Lab Assignment 3

Title: Implementing a real-time undo/redo system for a text editing application using a Stack data
structure. The system should support the following operations:
• Make a Change: A new change to the document is made.
• Undo Action: Revert the most recent change and store it for potential redo.
• Redo Action: Reapply the most recently undone action.
• Display Document State: Show the current state of the document after undoing or redoing an action

Program code:

```
#include <iostream>
#include <string>
using namespace std;

#define MAX 10

//Stack class
class Stack {
private:
    string arr[MAX];
    int top;

public:
    //Constructor
    Stack() {
        top = -1;
    }

    //Check if stack is empty
    bool isEmpty() {
        return top == -1;
    }

    //Check if stack is full
    bool isFull() {
        return top == MAX - 1;
    }

    //Push operation
    void push(string value) {
        if (isFull()) {
```

```
            cout << "Stack Overflow!\n";
        } else {
            top++;
            arr[top] = value;
        }
    }

    //Pop operation
    string pop() {
        if(isEmpty()) {
            cout << "Stack Underflow!\n";
            return "";
        } else {
            string value = arr[top];
            top--;
            return value;
        }
    }
};

//Main program
int main() {
    Stack undoStack;
    Stack redoStack;

    string document = "";
    string change;
    int choice;

    do {
        cout << "\n========Text Editor========\n";
        cout << "1. Make a Change\n";
        cout << "2. Undo Action\n";
        cout << "3. Redo Action\n";
        cout << "4. Display Document\n";
        cout << "5. Exit \n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch(choice) {
            case 1:
                cin.ignore();

                cout << "Enter text to add: ";
```

```
    getline(cin, change);

    //Save current document before changing
    undoStack.push(document);

    //Make new change
    document = document +change;

    //Clear redo stack
    while(!redoStack.isEmpty()) {
        redoStack.pop();
    }

    cout << "Change made successfully.\n";
    break;

case 2:
    if(undoStack.isEmpty()) {
        cout << "Nothing to undo!\n";
    } else {
        //Save current state for redo
        redoStack.push(document);

        //Restore previous state
        document = undoStack.pop();

        cout << "Undo successful\n";
    }
    break;

case 3:
    if(redoStack.isEmpty()) {
        cout << "Nothing to redo!";
    } else {
        //Save current state for undo
        undoStack.push(document);

        //Restore undone state
        document = redoStack.pop();

        cout << "Redo successful\n";
    }
    break;
```

```
            case 4:
                cout << "\nCurrent document: ";
                cout << document <<endl;
                break;

            case 5:
                cout << "Exiting program...\n";
                break;

            default:
                cout << "Invalid choice\n";
        }
    } while(choice != 5);

    return 0;
}
```

Output:

```
========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 1
Enter text to add: Hello
Change made successfully.

========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 1
Enter text to add:  World
Change made successfully.
```

========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 4

Current document: Hello World

========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 2
Undo successful

========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 4

Current document: Hello

========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 3
Redo successful

========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 4

Current document: Hello World

========Text Editor========
1. Make a Change
2. Undo Action
3. Redo Action
4. Display Document
5. Exit
Enter your choice: 5
Exiting program...